

Optimising Weak Reference Processing in HotSpot's Z Garbage Collector

Master's Thesis in Computer Science

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Agenda

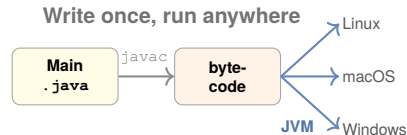
- 1 Background
- 2 Motivation
- 3 Design & Implementation
- 4 Evaluation Method
- 5 Results
- 6 Conclusions



Background

Java and the JVM

- ▶ One of the most used programming languages in the world
- ▶ Write once, compile once – runs on *any* machine via the **JVM**
- ▶ **HotSpot JVM** is the most popular implementation (Oracle / OpenJDK)



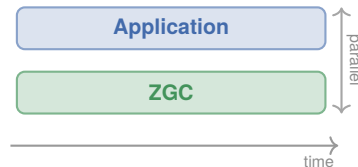
Garbage Collection (GC)

- ▶ Java has **automatic memory management** → no `free()` or `delete`
- ▶ The GC periodically finds objects the application can no longer reach and reclaims their memory

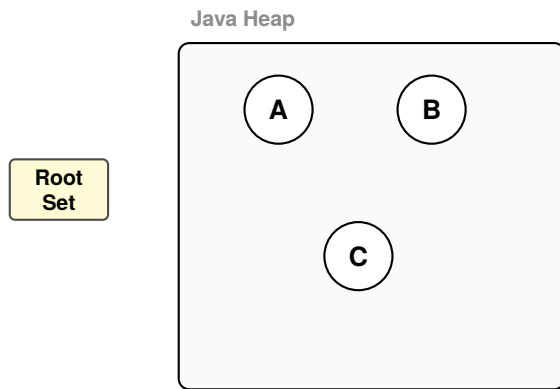
ZGC

- ▶ **Concurrent:** runs *alongside* the application → very short stop-the-world pauses
- ▶ **Relocating:** moves live objects to new compact regions. Old memory reused

ZGC runs concurrently

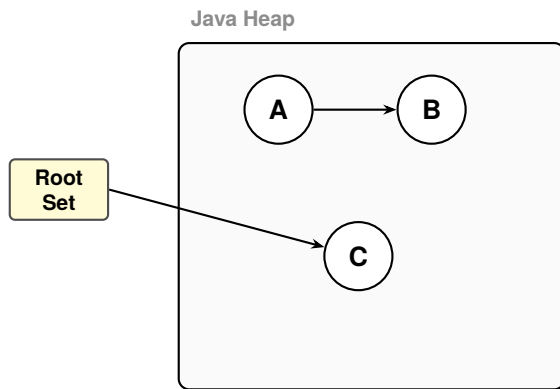


Garbage Collection



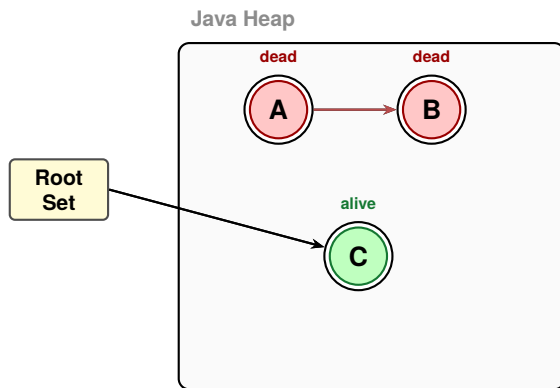
- Objects live on the **heap**, the root set is always alive

Garbage Collection



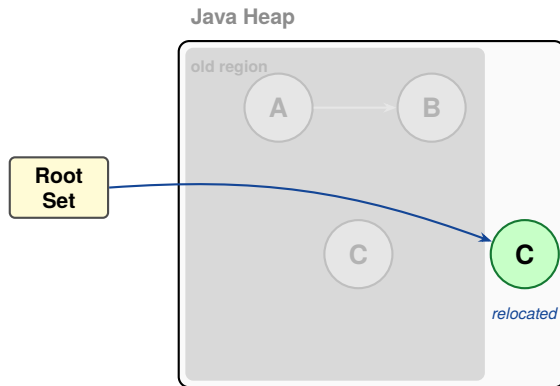
- ▶ Objects live on the **heap**, the root set is always alive
- ▶ GC **marks**: follows all references reachable from roots

Garbage Collection



- ▶ Objects live on the **heap**, the root set is always alive
- ▶ GC **marks**: follows all references reachable from roots
- ▶ **A** and **B** are unreachable → marked as **dead**. **C** is reachable → marked as **alive**

Garbage Collection



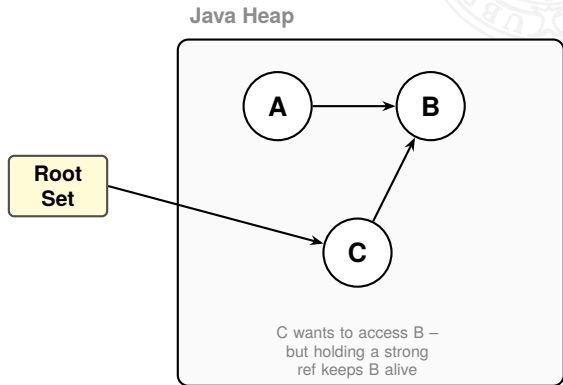
- ▶ Objects live on the **heap**, the root set is always alive
- ▶ GC **marks**: follows all references reachable from roots
- ▶ **A** and **B** are unreachable → marked as **dead**. **C** is reachable → marked as **alive**
- ▶ GC **relocates** C to a new region; old memory reused for future allocations

Weak References

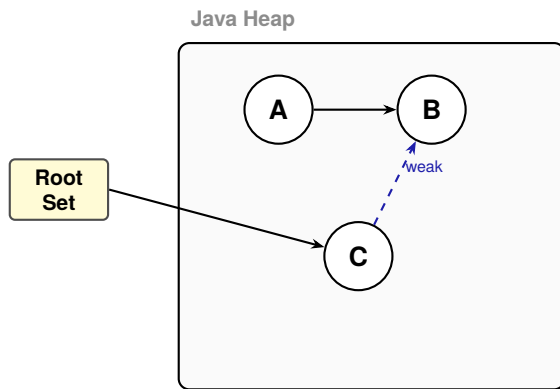
- ▶ Strong references keep objects alive. If you can reach it, it won't be collected
- ▶ But sometimes you want to **observe** an object *only while it is still alive*
- ▶ But if you hold a strong reference to the object, you keep it alive forever, defeating the purpose

The problem

You want to hold a reference that **does not** prevent garbage collection but still lets you read the object if it hasn't been collected yet.

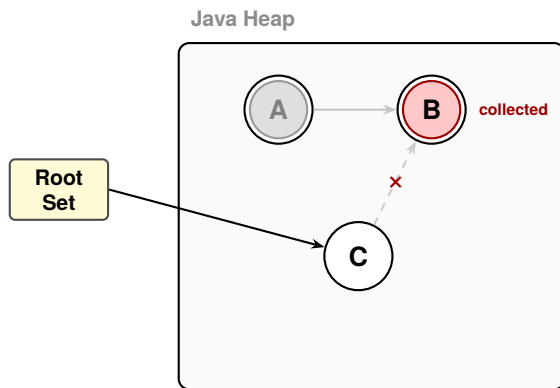


Weak References



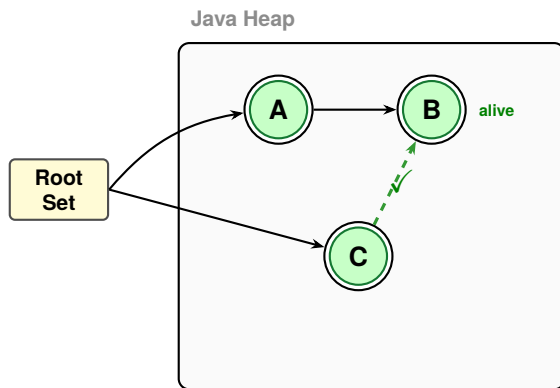
- ▶ A **weak reference** does not keep an object alive since it does not count as reachability for GC purposes

Weak References



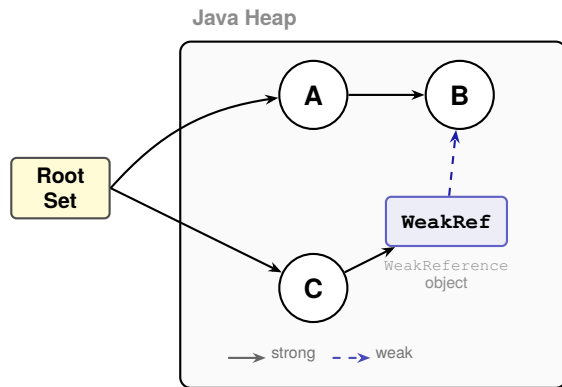
- ▶ A **weak reference** does not keep an object alive since it does not count as reachability for GC purposes
- ▶ If B has no *strong* path from roots: GC collects B; weak ref becomes `null`

Weak References



- ▶ A **weak reference** does not keep an object alive since it does not count as reachability for GC purposes
- ▶ If B has no *strong* path from roots: GC collects B; weak ref becomes `null`
- ▶ Add strong ref between root and A: B reachable through A → The weak ref from C to B is **valid**

The WeakReference Object



- ▶ The WeakReference object exists to support the ReferenceQueue callback
- ▶ When the GC collects B, the WeakReference is enqueued so the application can react

```
// With notification queue
new WeakReference<>(obj, queue)

// No queue
new WeakReference<>(obj)
```

Reference Processing Pipeline

1. Discovery Phase

Mark phase: `WeakReferences`
appended to a *discovered list*

2. Processing Phase

Referent alive? → keep
Referent dead? → clear it

3. Enqueue Phase

ReferenceHandler thread
delivers each cleared ref
to its `ReferenceQueue`



Motivation

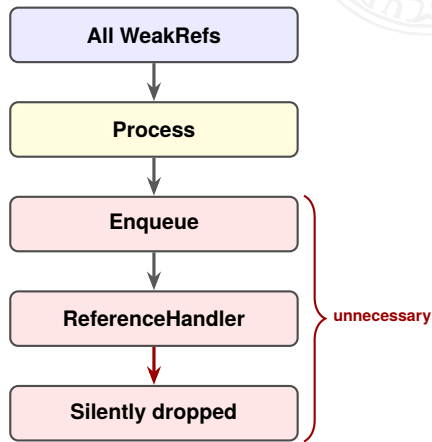
The Problem with Queue-less Weak References

- ▶ GC performance directly impacts application performance
- ▶ ZGC was designed for high performance – every unnecessary operation matters

The issue

When a `WeakReference` *without* a `ReferenceQueue` is cleared, it is still handed to the `ReferenceHandler` thread – which then silently drops it, since there is no queue.

Unnecessary work for every single queue-less weak reference.



Research Questions

RQ1 – Pipeline Optimisations

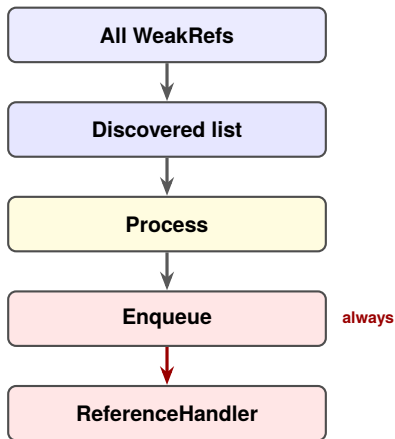
How do specific modifications to ZGC's processing of `WeakReference` objects *without* a `ReferenceQueue` affect reference-processing time, total GC cycle time, GCr memory footprint, and Java heap usage?

RQ2 – Weak Fields

How does expressing weak semantics through annotated object fields affect reference processing time, GC cycle time, GCr memory footprint, and Java heap usage compared to the optimised `WeakReference`-based approach of RQ1?

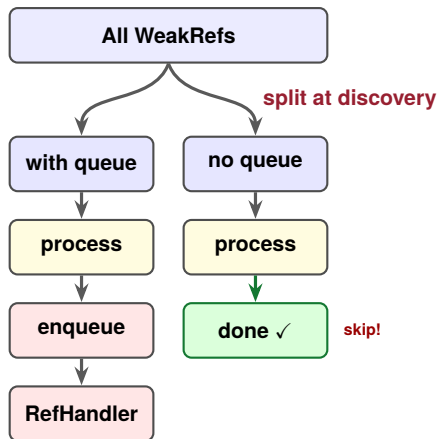
Design & Implementation

Optimisation 1: Skip Enqueue Path



- ▶ Every weak ref goes through the full pipeline
- ▶ Queue-less refs still hit the `ReferenceHandler` – which drops them
- ▶ Pure waste for weak references without a queue

Optimisation 1: Skip Enqueue Path

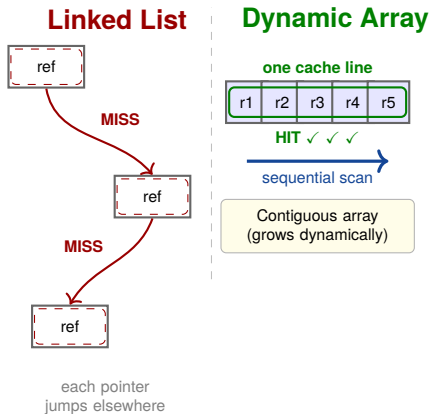


- ▶ At discovery: check the `ReferenceQueue` field
- ▶ Route queue-less refs to a *separate* discovered list
- ▶ No-queue pipeline: process (check + clear) – then **stop**
- ▶ With-queue: full existing behaviour

Why two lists instead of a branch?

A branch requires a memory load per ref to check the queue field. Two specialised loops avoid that per-reference cost.

Optimisation 2: Dynamic Array Discovered List

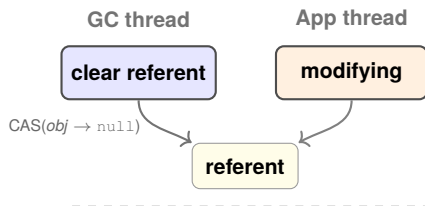


- ▶ Linked list: Nodes scattered across memory – no spatial locality
- ▶ Every node's next-pointer is likely a **cache miss**
- ▶ Dynamic Array: one cache load fills several consecutive entries
- ▶ Dynamically grow the array as needed

Trade-off

Higher auxiliary GC memory for array-based variants – explored in the Results section.

Optimisation 3: Optimised Clear Path



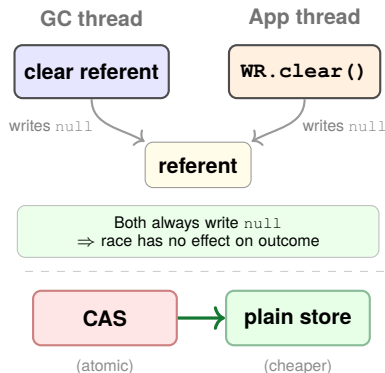
Compare-And-Swap (CAS):

```
if (*field == expected)
    *field = new_val; // atomic
```

Guarantees only one writer wins the race.

- ▶ Clearing a referent = writing `null` to the referent field
- ▶ The GC does this with a **CAS** – an atomic instruction that only writes if the current value matches an expected value
- ▶ Needed because the ZGC runs concurrently with the application – both may access the referent field at the same time without explicit synchronisation.

Optimisation 3: Optimised Clear Path



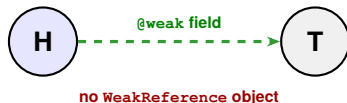
- ▶ **Key observation:** the only thing an app thread can write to a `WeakReference`'s referent field is `null` (via `WeakReference.clear()`)
- ▶ If a race occurs, both sides write `null` – the result is identical either way
- ▶ ⇒ The CAS is unnecessary; a **plain store** is safe

Research Question 2: Weak Fields (@weak)

WeakReference implementation



@weak field implementation



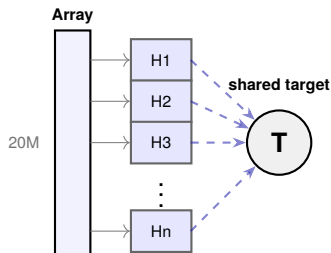
- ▶ A field annotated `@weak` is treated as a weak reference by ZGC – no `WeakReference` object needed
- ▶ Discovered during marking, stored in a dynamic array of weak fields
- ▶ Cleared the same way – but **cannot** use the optimised clear path (a regular field can be written to with any value, not just `null`)

```
class HolderClass {  
    @weak TargetClass ref;  
}
```

Evaluation Method

Benchmark Design

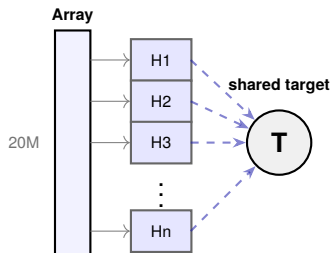
Single-Object Benchmark



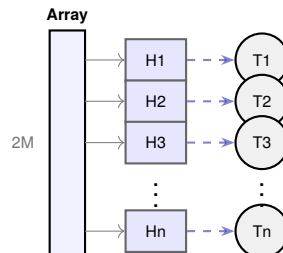
- **Single:** 20M holders, one shared payload – all 20M refs processed in one GC cycle

Benchmark Design

Single-Object Benchmark



Multi-Object Benchmark



- ▶ **Single:** 20M holders, one shared payload – all 20M refs processed in one GC cycle
- ▶ **Multi:** 2M holders, individual payloads, 5 gradual rounds – more realistic
- ▶ **Both:** no ReferenceQueue; both have a @weak field counterpart

Variants and Execution

Variant	clr	sep	dyn
None	—	—	—
Dyn	—	—	✓
Sep	—	✓	—
Sep + Dyn	—	✓	✓
Clear Path	✓	—	—
Clear Path + Dyn	✓	—	✓
Clear Path + Sep	✓	✓	—
All	✓	✓	✓
Weak Fields	—	✓	✓

- All $2^3 = 8$ pipeline combinations plus Weak Fields

Execution Environment

UPPMAX supercomputer (Pelle)

AMD EPYC 9454P (Zen 4)

48 physical cores, 768 GiB RAM

4 parallel instances per run

Each: 10 JVM cores + 2 aux cores

100 GB heap `-XX:+UseZGC`

250 iterations (+1 warmup discarded)

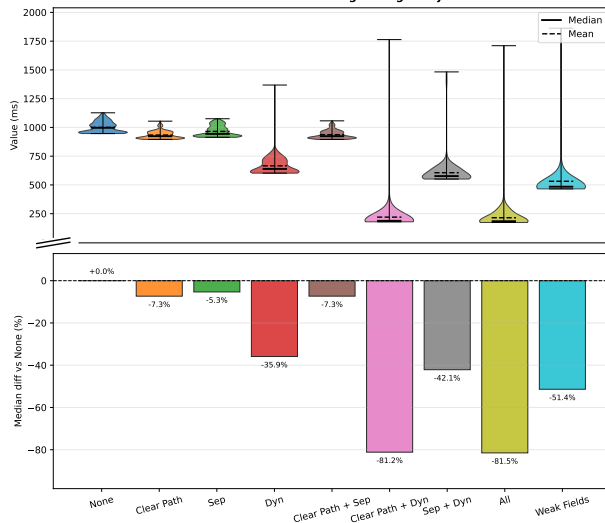
Reporting

Violin + **median** histogram over the 250 iterations.

Results

Non-Strong Processing Time – Single-Object Benchmark

Old Phase: Concurrent Process Non-Strong — Single-Object Benchmark



Variant

Reduction

Sep	−5%
Clear Path	−7%
Dyn	−36%
Clear Path + Dyn	−81%
All	−81%
Weak Fields	−51%

Super-additive interaction

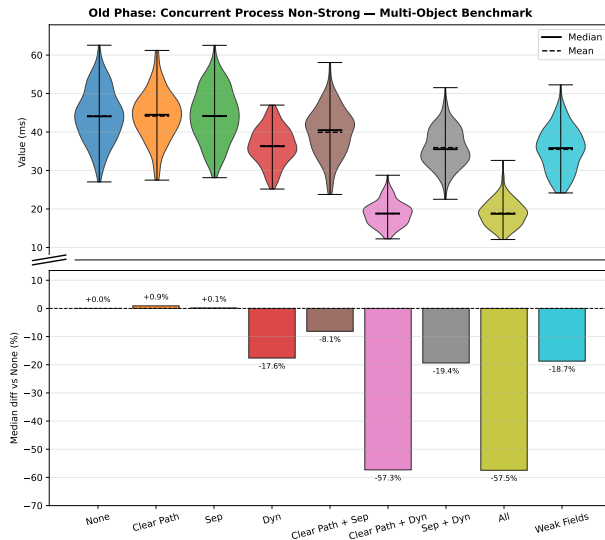
clear_path alone: -7%

dyn alone: -36%

Together: -81%

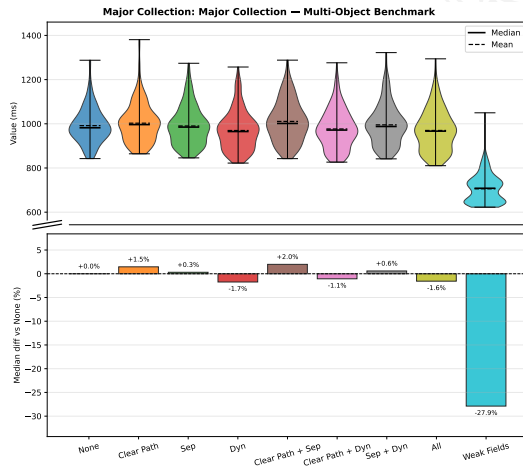
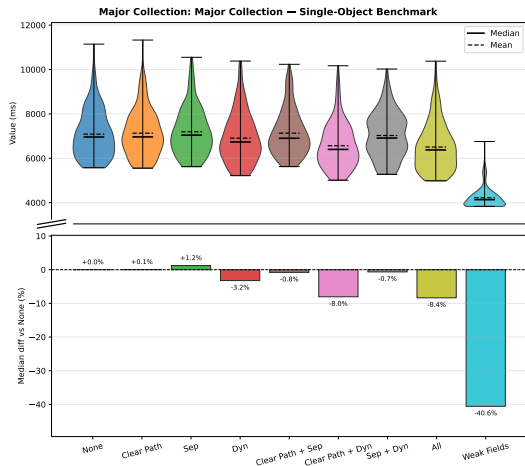
Each removes a different bottleneck.

Non-Strong Processing – Multi-Object Benchmark

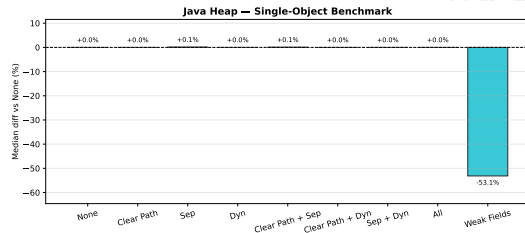
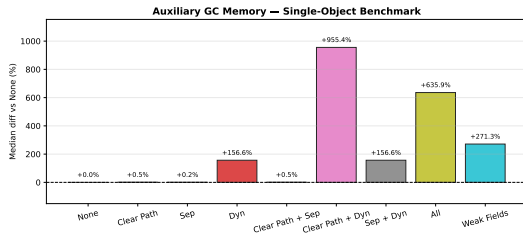


- ▶ Same pattern: Clear Path + Dyn and All at -57%
- ▶ Weak Fields: -19% – again comparable to Sep + Dyn

Major Collection Time



Memory Usage



- ▶ Clear Path + Dyn: $\times 10$ aux memory (1268 MB vs 120 MB)
- ▶ All: $\times 7$ (sep reduces bytes per entry vs dyn alone)

- ▶ All WeakReference variants: heap size identical
- ▶ Weak Fields: **-53%** heap (single), -5% (multi), no WeakReference objects to allocate

Conclusions

Conclusions – RQ1

RQ1 – Pipeline Optimisations

How do specific modifications to ZGC's processing of `WeakReference` objects *without* a `ReferenceQueue` affect reference-processing time, total GC cycle time, GC memory footprint, and Java heap usage?

- ▶ **81%** reduction in non-strong processing time (Clear Path + Dyn / All)
- ▶ Translates to only **8%** in total major collection time
- ▶ Hard ceiling: non-strong is 14.3% of major GC time (single) and 4.5% (multi)

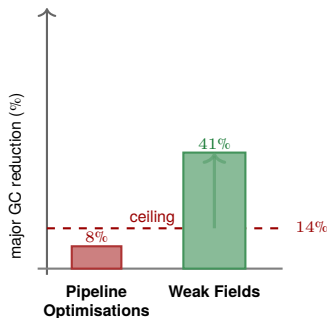
Conclusions – RQ2

RQ2 – Weak Fields

How does expressing weak semantics through annotated object fields affect reference processing time, GC cycle time, GC memory footprint, and Java heap usage compared to the optimised `WeakReference`-based approach of RQ1?

- ▶ **41%** major GC time reduction (single), **28%** (multi)
- ▶ **53%** heap reduction (single) – no `WeakReference` objects
- ▶ Structural advantage: fewer objects means less work across *every* GC phase

Key Takeaway



Optimise the pipeline (All)

-81% non-strong, but only -8% total GC

Hard ceiling: non-strong is just 14% of total GC

Change the representation (Weak Fields)

-41% total GC (single)

Fewer objects → less work in every GC phase

The bottleneck is the *representation*

Rethink how weak semantics is represented –
not just how to process `WeakReferences` faster.

Thank You

Questions?

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References I

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